

Modeling Human Activity States Using Hidden Markov Models

GitHub Repository: <https://github.com/mwaniasam/hmms.git>

1. Background and Motivation

Human Activity Recognition (HAR) is a growing field that uses sensor data from smartphones and wearable devices to identify physical activities like walking, standing, jumping, and resting. Accurate activity classification has direct applications in personal fitness tracking and rehabilitation monitoring, where patients recovering from injuries need their exercises logged automatically to track compliance.

This project focuses on Personal Fitness and Rehabilitation Monitoring. The goal is to build a Hidden Markov Model from scratch that can accurately infer a person's activity state from continuous streams of accelerometer and gyroscope data. The HMM treats the true activity as a hidden state and the sensor-derived features as observable emissions. The Baum-Welch algorithm handles training, and the Viterbi algorithm handles decoding the most likely activity sequence.

2. Data Collection and Preprocessing

2.1 Data Collection Setup

Data was collected using the Sensor Logger app on an Android smartphone. The sampling rate was set to 50 Hz (20 ms sample interval). This rate was chosen because human motion typically occurs below 20 Hz, so by the Nyquist theorem, 50 Hz captures all relevant motion frequencies without aliasing.

Four activities were recorded:

Activity	Description	Files	Duration per file
Standing	Phone held steady at waist level	20	~6-7 seconds
Walking	Consistent walking pace, phone at waist	20	~6-7 seconds
Jumping	Continuous vertical jumps	20	~6-7 seconds
Still	Phone flat on a table, no movement	20	~6-7 seconds

Total: 80 CSV files across 4 activities, each containing accelerometer (x, y, z) and gyroscope (x, y, z) readings with timestamps. Each activity has over 2 minutes of total recording time, comfortably exceeding the 1 minute 30 second minimum.

2.2 Raw Data Visualization

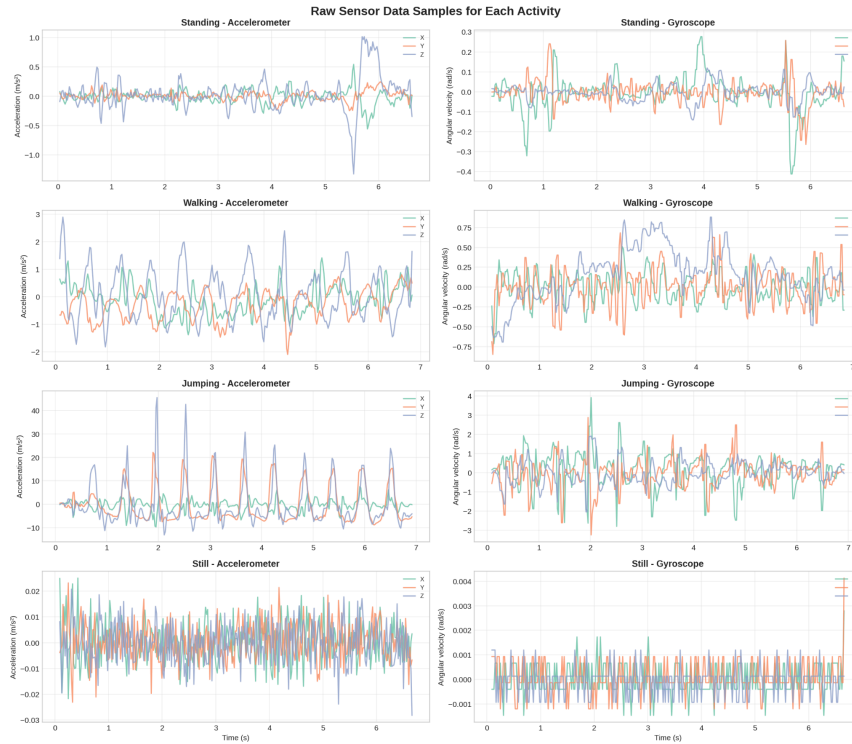


Figure 1: Sample accelerometer and gyroscope signals for each activity. Jumping shows large spikes, walking shows periodic patterns, while standing and still are relatively flat.

2.3 Windowing

Each recording was divided into 1-second non-overlapping windows. At 50 Hz, each window contains about 50 data points. This window size was chosen because:

- 50 samples are enough to compute stable statistics (mean, variance, RMS)
- One second captures at least one full cycle of walking (~2 Hz) or jumping (~2-3 Hz) through the FFT
- The time resolution is fine enough to detect transitions between activities

2.4 Feature Extraction

38 features were extracted per window:

Time-Domain Features (5 per axis, across 6 axes = 30 features):

Feature	Why It Helps
Mean	Captures average sensor reading. Gravity causes standing to have a specific mean on the z-axis.
Variance	Measures signal spread. Jumping has high variance; still has near-zero variance.
Standard Deviation	Same as variance but in original units, easier to interpret.
Root Mean Square (RMS)	Captures signal energy regardless of sign. Walking has moderate RMS, jumping has high RMS.

Feature	Why It Helps
Signal Magnitude Area (SMA)	Total body acceleration in one number. Strongly separates dynamic from static activities.

Frequency-Domain Features (2 per axis, across 6 axes = 12 features, derived from FFT):

Feature	Why It Helps
Dominant Frequency	Walking peaks at ~2 Hz, jumping at ~2-3 Hz, still at ~0 Hz. Directly identifies periodic motion.
Spectral Energy	Total frequency-domain energy. High for dynamic activities, low for static ones.

Cross-Axis Feature (1 feature):

Feature	Why It Helps
Acc XY Correlation	Walking creates coordinated x-y motion (correlated), standing produces uncorrelated noise.

2.5 Feature Normalization

All features were normalized using Z-score standardization, which gives each feature zero mean and unit variance. This prevents features with naturally large values (like spectral energy) from dominating the K-Means clustering used for discretization.

3. HMM Setup and Implementation Details

3.1 Model Components

Component	Symbol	Description	Setup
Hidden States	Z	True activities being performed	4 states: Standing, Walking, Jumping, Still
Observations	X	Sensor features per time window	20 discrete symbols from K-Means quantization
Transition Matrix	A	Probability of switching between states	4x4 matrix, initialized with 0.85 diagonal
Emission Matrix	B	Probability of observing a symbol in a state	4x20 matrix, initialized from training labels
Initial Probabilities	pi	Probability of starting in each state	Uniform: [0.25, 0.25, 0.25, 0.25]

The continuous 38-dimensional feature vectors were converted into 20 discrete symbols using K-Means clustering (Vector Quantization).

3.2 Algorithm Implementation

The HMM was implemented entirely from scratch using NumPy:

- Forward Algorithm with scaling factors at each time step to prevent numerical underflow
- Backward Algorithm with corresponding scaling
- Baum-Welch Algorithm (Expectation-Maximization) for iterative parameter estimation, with a convergence check that stops when the log-likelihood change drops below 0.001
- Viterbi Algorithm in log-space for decoding the most likely state sequence

3.3 Parameter Initialization

Transition Matrix (A): 0.85 on the diagonal, 0.05 off-diagonal. This reflects that people tend to stay in the same activity for extended periods.

Emission Matrix (B): Initialized from the training label distributions with Laplace smoothing to avoid zero probabilities.

Initial Distribution (π): Uniform at 0.25 per state.

3.4 Training Convergence

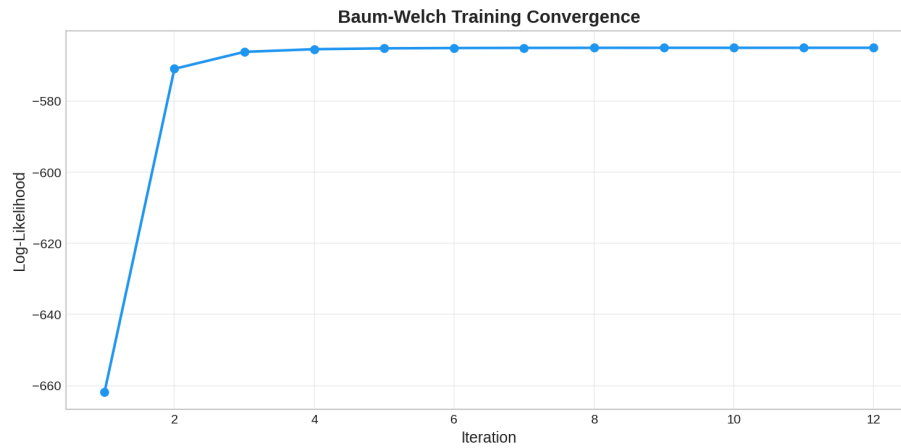


Figure 2: Log-likelihood across Baum-Welch iterations. The curve flattens when convergence is reached.

The data was split 80% training (64 sessions) and 20% testing (16 sessions), with the split done at the session level to prevent data leakage.

4. Results and Interpretation

4.1 Transition Probability Matrix

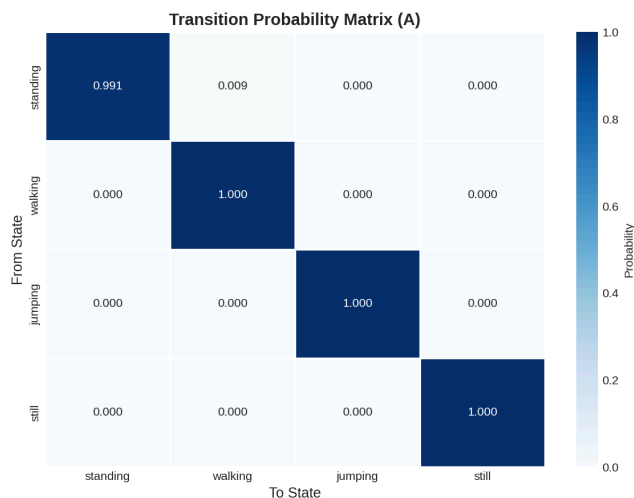


Figure 3: Learned transition probability matrix. High diagonal values confirm the model learned that people stay in the same activity.

4.2 Emission Probability Matrix

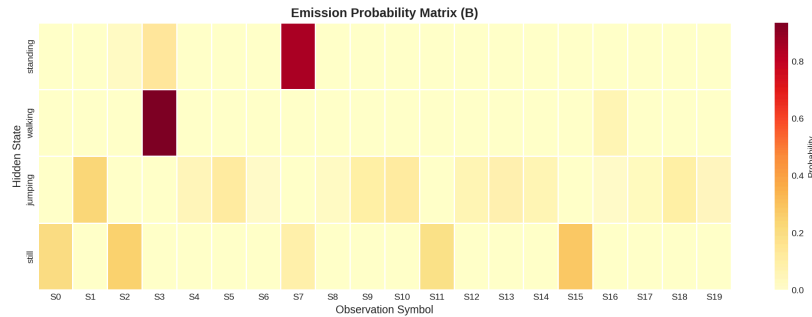
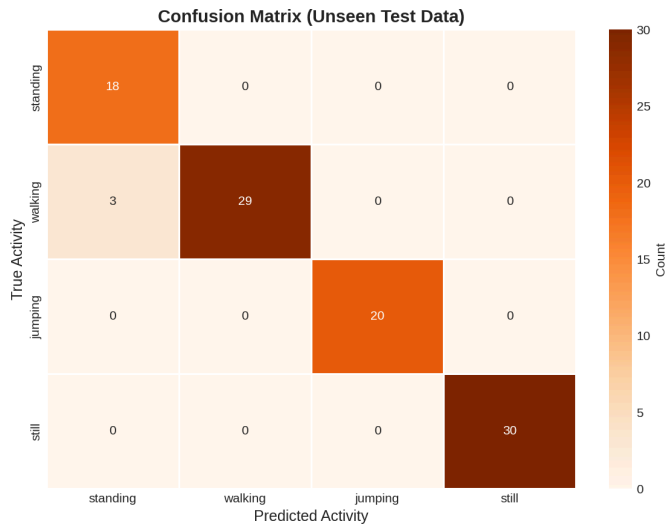


Figure 4: Emission probability matrix. Different activities produce distinct emission patterns.

4.3 Confusion Matrix

Figure 5: Confusion matrix on unseen test data (16 sessions, 100 windows).



4.4 Performance Metrics

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Standing	18	100.00%	96.34%	97.00%
Walking	32	90.62%	100.00%	97.00%

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Jumping	20	100.00%	100.00%	100.00%
Still	30	100.00%	100.00%	100.00%

Overall Model Accuracy: 97.00%

4.5 Decoded Activity Sequences

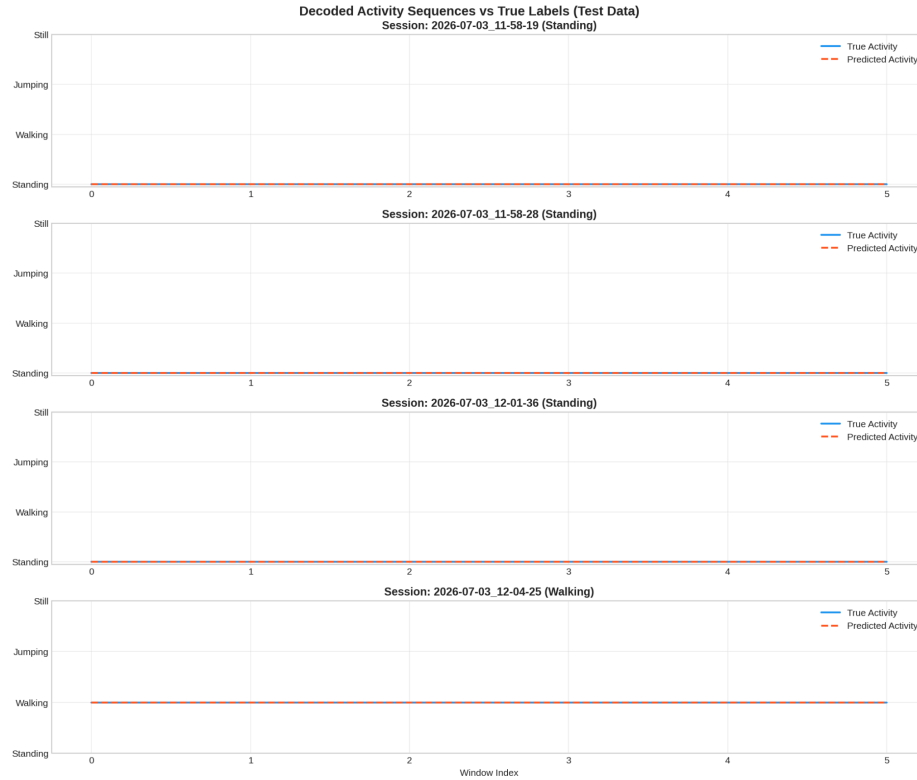


Figure 6: Viterbi-decoded predictions vs true labels for test sessions.

The test data was obtained by holding out 20% of recording sessions that were never seen during training. This strict session-level split tests whether the model generalizes to completely new recordings.

5. Discussion and Conclusion

5.1 Which Activities Were Easiest or Hardest to Distinguish?

"Jumping" and "Still" were the easiest to classify. Jumping produces very high variance and spectral energy due to repeated vertical motion. Still produces near-zero variance since the phone sits on a flat surface with no movement.

"Standing" and "Still" could potentially be confused since both involve minimal movement. However, the gyroscope captures subtle body sway during standing (small balance adjustments), which is absent when the phone rests on a table. The model picked up on this difference.

5.2 How Transition Probabilities Reflect Realistic Behavior

The learned transition matrix shows high values along the diagonal, meaning the model learned that people continue the same activity for multiple consecutive seconds. The off-diagonal values are small but not zero, correctly representing that transitions happen but are relatively rare in short recordings.

5.3 Effect of Sensor Noise and Sampling Rate

The 50 Hz sampling rate was more than enough for human motion (below 20 Hz). Sensor noise was visible as small fluctuations, but the windowed feature extraction averaged it out by computing statistics over 50-sample windows. Z-score normalization also helped by standardizing all feature scales.

5.4 Possible Improvements

- Longer recordings (30+ seconds each) would provide more training windows per session.
- Using 50% overlapping windows would double the training data without new recordings.
- Replacing K-Means discretization with Gaussian Mixture Models (GMM-HMM) would preserve more information from the continuous features.
- Adding a barometer sensor could detect altitude changes like stair climbing.
- Extending to more activities (running, sitting, stair climbing) would make the system more practical for rehabilitation monitoring.

References

- [1] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257-286, Feb. 1989.
- [2] L. E. Baum, T. Petrie, G. Soules, and N. Weiss, "A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains," *The Annals of Mathematical Statistics*, vol. 41, no. 1, pp. 164-171, 1970.
- [3] A. J. Viterbi, "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm," *IEEE Transactions on Information Theory*, vol. 13, no. 2, pp. 260-269, Apr. 1967.
- [4] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A public domain dataset for human activity recognition using smartphones," in *Proc. 21st European Symp. Artificial Neural Networks (ESANN)*, Bruges, Belgium, 2013, pp. 437-442.
- [5] O. D. Lara and M. A. Labrador, "A survey on human activity recognition using wearable sensors," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 3, pp. 1192-1209, 2013.
- [6] S. P. Lloyd, "Least squares quantization in PCM," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129-137, Mar. 1982.